Speed Control of BLDC Motor: Design and Simulation Based on Neural Network

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***Abstract*— Traditional speed control methods for BLDC motors often rely on conventional control algorithms, which may have limitations in terms of accuracy and adaptability. The neural network is designed to learn the nonlinear characteristics of the BLDC motor and adaptively adjust the control signals to achieve precise speed control. The neural network architecture is carefully constructed to capture the complex relationships between the motor inputs and outputs. The simulation environment is established using MatLab/SIMULINK, and the proposed neural network-based speed control system is implemented and tested. System performance is evaluated under various operating conditions, including different speed references. The simulation results demonstrate the effectiveness of the proposed neural network-based speed control system. The system exhibits superior performance in terms of accuracy, robustness, and adaptability compared to traditional control methods. It successfully maintains the desired speed in the presence of disturbances and load variations, showcasing the neural network's ability to learn and adapt. The results of this study expand BLDC motor control methods and may spur other advancements in the area of neural network-based control systems.**

Keywords—NNA, BLDC, Speed Control, Three Phase Inverter, MatLab Simulink, PI.

I. Introduction

Owing to their high efficiency, small size, and low maintenance requirements, brushless DC (BLDC) motors have become popular in a variety of industrial applications. However, for the BLDC motors to operate at their best, precise speed control must be achieved. Traditional algorithms, including proportional-integral-derivative (PID) controllers, are often used in BLDC motor control systems. The precision and flexibility of these approaches may be constrained in the presence of nonlinearities and disturbances, even though they may still offer sufficient control[1]–[4].

In many industrial applications, the design and modeling of speed management of Brushless DC (BLDC) motors is crucial. BLDC motors are often employed in robots, electric cars, and renewable energy systems, owing to their many benefits, including high efficiency, small size, and minimal maintenance requirements. For these applications, achieving accurate speed control is essential for maximizing the performance[5]-[11].

Researchers have been investigating the use of neural networks as an alternative method for speed control of BLDC motors to overcome these restrictions. Neural networks are computer models that are inspired by the structure and operation of the human brain. They can adjust to changing environments and learn complex correlations between inputs and outcomes [12]–[17].

In this paper, we propose a neural-network-based speed control system for BLDC motors, aiming to design a system that can accurately control motor speed while being robust to disturbances and load variations. The neural network learns the nonlinear characteristics of the motor and generates control signals based on input parameters, such as the motor voltage, current, and rotor position. The design and simulation of the proposed system will be performed using a suitable software tool, replicating real-world operating conditions to evaluate its performance. By leveraging the learning capabilities of neural networks, this study aimed to overcome the limitations of conventional control methods and advance the field of BLDC motor control.

II. LITERATURE REVIEW

This section presents a literature review of the design and simulation speed control of a BLDC motor based on a neural network.

In 2020, T. A. Shifat and H. Jang-Wook, [18] proposed a deep learning-based approach to evaluate the residual useful life (RUL) of BLDC motors affected by stator-related issues. There are two distinct types of stator problems: inter-turn faults (ITF) and winding short-circuit faults (WSC). The method uses a recurrent neural network (RNN) and an average-value rectifier (AVR) for voltage degradation data. To estimate nonlinear trends, long-short-term memory (LSTM) with an attention mechanism was used. The suggested approach is appropriate for BLDC motor health status estimation and real-time condition monitoring, and is

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not influenced by external factors, making it appropriate for a variety of operating circumstances.

In 2022, S. Kanagaraj and S. Nithaiyan, [19] suggested a current and speed management strategy to reduce torque ripple in BLDC motors in hybrid power systems for renewable energy sources. This technique increases dependability and flexibility in system error by optimizing the speed and error torque using the mayfly optimization algorithm (MOA). The approach is put into practice in MATLAB/Simulink and contrasted with previously studied methods, such as the moth flame method (MFA) and particle swarm optimization (PSO).

In 2023, D. K. Shary et al, [20] in this study, the control of brushless DC motor speed in the industrial and automotive industries is examined. It creates a mathematical model and evaluates its speed using the classic PID, particle swarm optimization (PSO), and neural network PID controller kinds of controllers. The first controller, according to the results, lowers speed to 1,526 rpm while the second and third controllers do away with oscillations.

In 2018, A. H. Ahmed et al, [21] In this work, the dynamic performance of fuzzy logic controllers (FLC) and proportional-integral (PI) motor drives were compared with brushless DC motor drives. FLC is more suitable for load disturbances than PI, despite PI's drawbacks. BLDC motors' dynamic properties are compared using MATLAB/SIMULINK modeling and simulation under various operating situations.

In 2020, J. N. Bae et al, [22] This study proposed a 2-pole high-speed BLDC motor design for kitchen blenders that reduces speed ripple and improves grinding efficacy—reduced speed ripple results from the motor's constant modification of its speed reference to satisfy the maximum pulsed torque requirements. In trials and simulations, hard white beans are utilized to assess how well the control strategy performs. Computations and experiments verify the design results and grinding performance.

In 2021, B. Azhari et al, [23] created a BLDC motor with a 5 kW, 100 volt design utilizing a stator coupled to an inverter with Hall sensors. The design considers inverter characteristics, winding layout, and rotor elements. Analytical calculations and finite element magnetic simulation demonstrate 5.1 kW output power, 97.09% efficiency, and 32.82 Nm torque. The motor operates with the attached inverter, demonstrating its effectiveness.

In 2023, H. S. Dakheel et al, [24] This article discussed how the brushless direct current (BLDC) motor is a popular option for industrial applications due to its high torque. One kind of controller that is used to improve performance is the proportional integral derivative or PID controller. An appropriate set of parameters for kp, ki, and kd are obtained by tuning the PID controller using a genetic algorithm (GA). The GA, conventional control units, and the error control standard for integral time absolute error (ITAE) type errors were all used in the simulations. The findings demonstrated the advantages of GA over conventional controllers in terms of reaction time, stability, increase and settling time, and percentage overshoots.

In 2023, Q. D. Nguyen et al, [25] emphasized that this study is focused on the low cost, high torque, and performance of a speed control system for brushless DC motors (BLDC) in smart storage cabinets. Cabinet movement was controlled using a proportional-integral (PI) controller. It will analyze the system's construction and tune the Ziegler-Nichols technique and operating principle. Utilizing an STM32 microcontroller unit, the experimental system was set up. The power of the proposed PI controller was shown in both no-load and load situations, exhibiting its efficiency in the following speed signals.

In 2016, Suneeta et al, [26] outlined a technique for creating and managing affordable and effective Brushless Direct Current (BLDC) motors. An algorithm was created utilizing the resolver signals recorded from the motor and FPGA characteristics. The Spartan-3 FPGA device was subsequently loaded with the MOSFET firing pulses generated by the program. Comparing the hardware and programmed PIC microcontroller-based systems, it was discovered that they are more effective and dependable.

In 2021, I. Etier et al, [27] designed and examined the performance of electronically controlled brushless direct current (BLDC) motors in MATLAB. The BLDC motor is extensively used in several industries, and it requires a less ripple-producing input supply and a decent speed controller. The optimum performance is found with a three-phase VSI-fed BLDC motor and FOPID controller, according to a study that examined the torque, voltage, and speed response.

III. Neural Network Algorithm (Extreme Learning Machines (ELM))

Extreme Learning Machines are a type of learning for single-hidden-layer feedforward neural networks (ELM). It offers rapid learning by randomly allocating the weights between the input and hidden levels and analytically determining the weights between the hidden and output layers. A BLDC motor with Hall sensors and speed inputs will be adaptively controlled by ELM, as seen in Fig. 1.

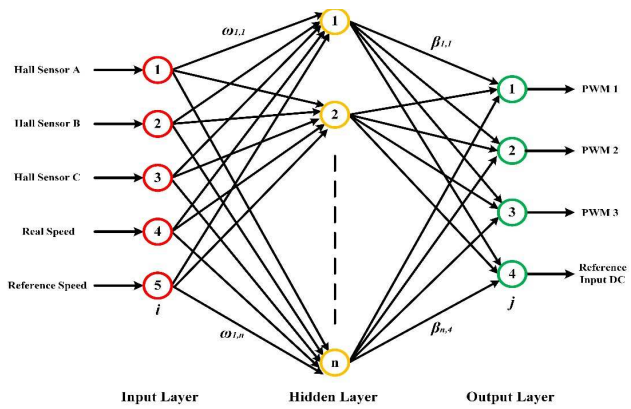

Fig. 1. Neural network Algorithm for the BLDC motor speed control

The ELM network consisted of an input layer, an output layer, and one hidden layer. The output layer sends the control signal to the motor, whereas the input layer receives the speed and Hall sensor input. Weights and biases were randomly allocated to each buried layer. The BLDC motor was driven by a set of input signals in order to collect training data. Appropriate Hall sensor and speed measurements were recorded together with the intended control signal.

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The hidden layer activation values were calculated by passing the input data through randomly assigned weights and biases of the hidden layer. The activation values can be obtained using a nonlinear activation function, such as a radial basis function or sigmoid function. The output weights were computed by solving a linear equation that maps the hidden layer outputs to the desired control signal. The control signal for the BLDC motor was estimated by multiplying the output weights by the hidden layer activation values.

According to Fig. 1, the following equations are used to express the ELM in Eq. One for the activation function f(x), hidden neurons H, and training data N [28]. Linear Eq. Two and Eq.3 were used to compute the output weights.

$$Y_j = \sum_{i=1}^{H} \beta_i f(\omega_i, c_i, x_i) \cdot j = 1 \dots \dots . N \qquad (1)$$

Where

Y_j: Output from ELM:: ω_i: Input weights : β_i: Output weights :: x_i: Input variables: c_i: Hidden bias

$$\beta = A \dagger Y \qquad (2)$$

Where

The Moore-Penrose generalized inverse of A is represented as $A \dagger$, and Y.

$$A = \begin{vmatrix} h(x_i) \\ \vdots \\ h(x_N) \end{vmatrix} = \begin{vmatrix} f(\omega_1, c_1, x_1) & \dots & f(\omega_H, c_H, x_1) \\ \vdots & \ddots & \vdots \\ f(\omega_1, c_1, x_j) & \dots & f(\omega_H, c_H, x_J) \end{vmatrix} \qquad (3)$$

Where

$h(x_i)$:Is the hidden layer feature mapping

IV. RESULTS AND DISCUSSION

This section clarifies the research findings and provides a detailed analysis. The results of the research can be divided into two parts, which are the results of the three-phase inverter, as well as the BLDC motor's outcomes With MATLAB/Simulink, the BLDC motor and the finished system are simulated, Table I provides a summary of the BLDC motor's simulation-related parameters. The block diagram in Fig. 2 shows the final control system using an ANN algorithm.

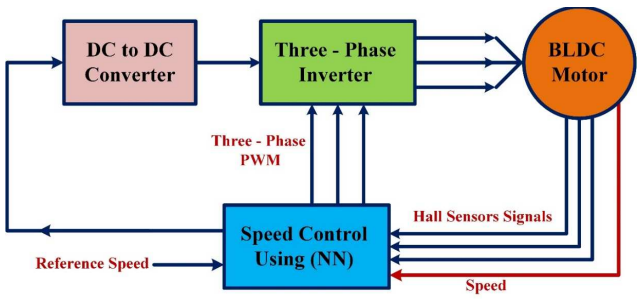

Fig. 2. Block diagram for the final proposed system of the BLDC motor speed control

TABLE I. MAIN PARAMETERS OF BLDC MOTOR

Parameter	Value
Number of phases	3
Stator phase resistance Rs	0.7 (Ohm)
Stator phase inductance Ls	2.72e-3 (H)
Flux linkage	0.105
Back EMF flat area	120 (degrees)
Rated speed and Rated torque	1650 (rpm) /1.5(Nm)

A. *Three-phase inverter results*

When using an ANN to run Three-Phase BLDC motors under speed control, according to Fig. 2, it is necessary to be aware of the output for each part under speed control. Fig. 3 shows the DC link voltage applied to the terminals of the three-phase inverter during the change in reference speeds.

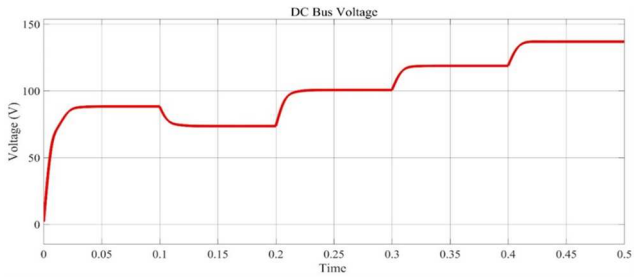

Fig. 3. DC Link voltage For Three-Phase inverter

By reviewing Fig. 4, we notice how the pulse width of the six switches is changed to control the inverter output results for speed control under the ANN algorithm. With a change in the reference speed, the PWM is also changed, as shown in Fig. 4, which causes a change in the three-phase line-to-line output voltage of the inverter, as shown in Fig.5.

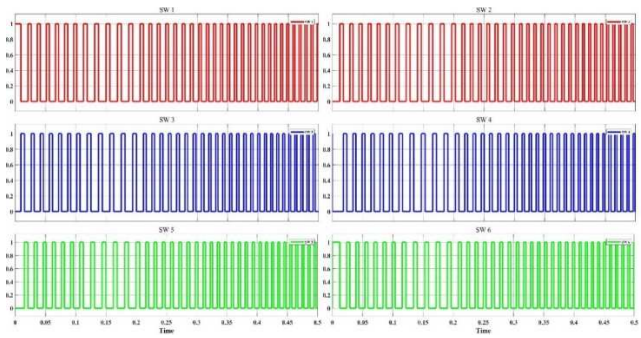

Fig. 4. Six PWM for a three-phase inverter under ANN control system

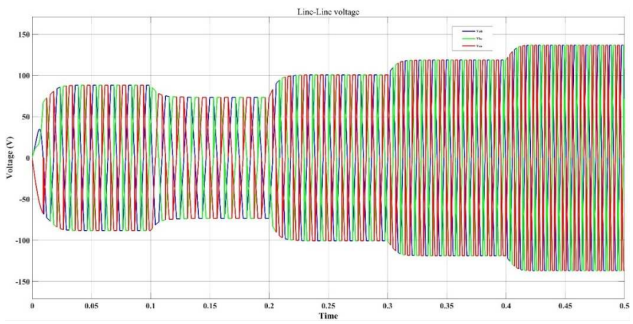

Fig. 5. The output voltage for three phase inverter (Line-to-Line Voltage)

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B. BLDC Motor Results

To operate the motor constantly under the ANN, as shown in Fig. 2, the output of the inverter must match the position of the rotor at any given time. The three Hall sensors may provide six potential states, and the inverter switching state corresponds to each of these states. The rotor's present location is determined by the states of the Hall sensors, which provide the data required for electronic commutation. An example of the Hall sensor output with the motor running is shown in Fig. 6.

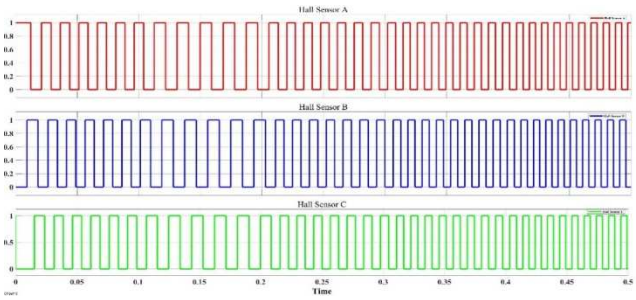

Fig. 6. Hall sensors results for BLDC motor

The response accuracy of the BLDC motor model was determined by running it with the tuned system control parameters obtained and performing an output plot analysis. The output of the automated model for the system control parameters and replicated close motor speed are shown in Fig. 7. The desired speeds were entered into the model with RPM values of 1000, 800, 1100, 1300, and 1500. These values are represented in red, whereas those for the control reaction are shown in blue. The required speed values were specified, and the system control reaction followed them. The tuned system controller was successful in managing the speed of the model in line with the specified intended speed because the system response adhered to the supplied speed, even though the speed fluctuated continually.

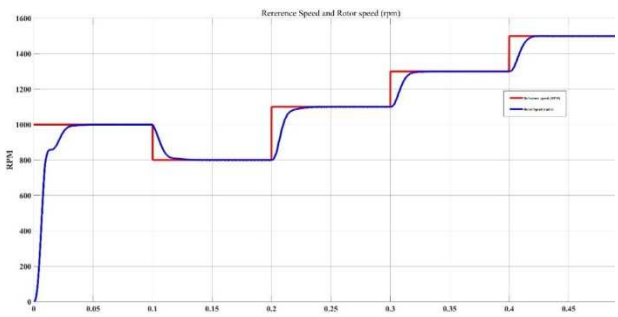

Fig. 7. The speed response of a BLDC motor under different reference speed settings (Stator of current BLDC motor under ANN control system).

The information in the passage is mostly about looking at how the stator current and torque change when the system's speed changes continuously, as shown in Figs. 8 and 9.

Fig. 8 shows the stator current's behavior with system speed changes, providing crucial information about the system's dynamics. The graph reveals that the stator current varies with system speed variations, which is essential for assessing motor efficiency and understanding how speed variations impact it.

Fig. 9 focuses on torque fluctuations that coincide with speed variations, an essential motor component that influences system efficiency and rotational motion. The graph of torque response to continuous speed variations provides insight into system dynamics, torque levels over time, and system stability. A thorough analysis of torque response is necessary to ensure the engine's stability and flexibility in changing speed conditions.

When the data from Figs. 8 and 9 are integrated, a thorough depiction of the system behavior under continuous velocity changes is produced. A comprehensive picture of the dynamic reaction of the motor at different speeds is provided by stator current and torque ratings. For engineers and researchers who work to optimize systems stability and performance, it allows them to mark patterns, trends, and potential areas for advancing the motor control technique.

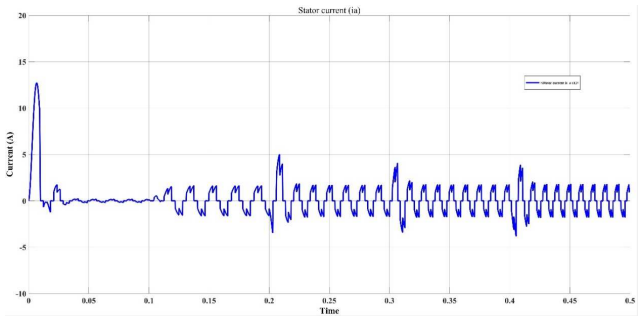

Fig. 8. Stator current of BLDC motor under ANN control

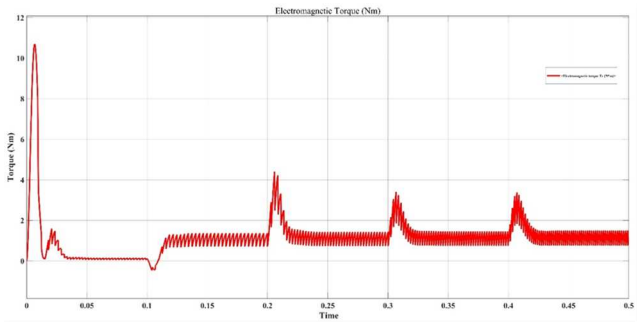

Fig. 9. The electromagnetic torque of BLDC motor is under ANN control

V. CONCLUSION

Positive results have been obtained from neural network-based modeling and BLDC motor speed regulation design. By understanding the compound link between motor inputs and the output of intended speed, the neural network exhibits an incredible ability to manage the motor speed correctly. The neural network-based system performed utmost speed control while matching traditional control methods. Its nonlinearity and system dynamics enable more precise control and faster response

One benefit of a neural network-based speed control system is its capability to adjust to changing conditions of operating. In the face of changing conditions, the network's ability to generalize and adapt is critical to maintaining stable control performance. Moreover, the neural network-based approach showed low sensitivity to changes in kinetic

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parameters, a useful property for real-world applications. Since less precise data is needed for engine parameters, this feature improves system reliability and ease of deployment.

Ultimately, the accuracy and flexibility of BLDC speed control were promoted through neural network simulation and design, opening the door for further study and real-world application of neural network-based control techniques.

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